习题 2.1

(1) $A=\begin{bmatrix} l+x & xy \\ xy & -y^2 \end{bmatrix}$ ~~双曲型~~

$$\lambda^2-(l+x-y^2)\lambda-x^2y^2-(l+x)y^2=0$$

~~双曲型方程~~，区域 $\{(x,y)\mid x,y\neq 0\}$. 区域

~~抛物型方程，区域 $\{(x,y)\mid (x=0\wedge[(l\geq 0\wedge y\neq\sqrt{l})\vee(l<0)])\vee(y=0\wedge x\neq -l)\}$.~~

~~椭圆型方程：~~

椭圆 ~~双曲~~型：$\{(x,y)\mid y\neq 0\wedge(x^2+x+l<0)\}\Leftrightarrow\{(x,y)\mid (y\neq 0)\wedge(l<\frac{1}{4})\wedge(-\frac{1}{2}-\sqrt{\frac{1}{4}-l}<x<-\frac{1}{2}+\sqrt{\frac{1}{4}-l})\}$

抛物型：$\{(x,y)\mid (y=0\wedge x\neq -l)\vee[(l\leq\frac{1}{4})\wedge(x=-\frac{1}{2}\pm\sqrt{\frac{1}{4}-l})\wedge(y^2\neq x+l)\wedge(y\neq 0)]\}$

双曲 ~~椭圆~~型：$\{(x,y)\mid (x^2+x+l)y^2>0\}=\{(x,y)\mid y\neq 0\wedge[(l>\frac{1}{4})\vee(x>-\frac{1}{2}+\sqrt{\frac{1}{4}-l})\vee(x<-\frac{1}{2}-\sqrt{\frac{1}{4}-l})]\}$

习题 2.2

$$\begin{cases}\partial_x=\partial_\xi\xi_x+\partial_\tau\tau_x\\ \partial_t=\partial_\xi\xi_t+\partial_\tau\tau_t\end{cases}\Rightarrow\begin{cases}u_x=\dfrac{\partial u}{\partial\xi}\\ u_t=\dfrac{\partial u}{\partial\xi}\cdot(-\alpha)+\dfrac{\partial u}{\partial\tau}\end{cases}\Rightarrow u_{xx}=\frac{\partial^2u}{\partial\xi^2}$$

$$\Rightarrow\ \partial_t u+\alpha\partial_x u=\alpha^2\partial_{xx}u\ \Leftrightarrow\ -\alpha\partial_\xi u+\partial_\tau u+\alpha\partial_\xi u=\alpha^2\partial_{\xi\xi}u$$

$$\Leftrightarrow\ \partial_\tau u=\alpha^2\partial_{\xi\xi}u$$

习题 2.3

(1) $\partial_x=\partial_\xi+\partial_\eta$，$\partial_t=a(\partial_\xi-\partial_\eta)$

$$\partial_{xx}=(\partial_\xi+\partial_\eta)^2=\partial_{\xi\xi}+2\partial_{\xi\eta}+\partial_{\eta\eta}$$

$$\partial_{tt}=a^2(\partial_{\xi\xi}-2\partial_{\xi\eta}+\partial_{\eta\eta})$$

$$0=\partial_{tt}u-a^2\partial_{xx}u=-4a^2\partial_{\xi\eta}u\Rightarrow\partial_{\xi\eta}u=0\Rightarrow\frac{\partial^2u}{\partial\xi\partial\eta}=0$$

$$\Rightarrow\frac{\partial u}{\partial\eta}=f(\eta)\Rightarrow u=\int_{\eta_0}^{\eta}f(\eta)\,d\eta+g(\xi)$$

令 $g(\xi)=F(\xi)=F(x+at)$，$\int_{\eta_0}^{\eta}f(\eta)\,d\eta=G(\eta)=G(x-at)$

$$\Rightarrow u(x,t)=F(x+at)+G(x-at)$$

(2)
$$\begin{cases}\varphi(x)=F(x)+G(x)\\ \psi(x)=aF'(x)-aG'(x)\end{cases}\Rightarrow\begin{cases}F(x)+G(x)=\varphi(x)\\ F(x)-G(x)=\dfrac{1}{a}\displaystyle\int_0^x\psi(\theta)\,d\theta+C\end{cases}\Rightarrow\begin{cases}F(x)=\dfrac{1}{2}\left(\varphi(x)+\dfrac{1}{a}\displaystyle\int_0^x\psi(\theta)\,d\theta+C\right)\\ G(x)=\dfrac{1}{2}\left(\varphi(x)-\dfrac{1}{a}\displaystyle\int_0^x\psi(\theta)\,d\theta-C\right)\end{cases}$$

$u(x,t) = \frac{1}{2}(\varphi(x+at) + \varphi(x-at)) + \frac{1}{a}\int_{x-at} \psi(\theta)d\theta$

習題 2.4

(1) 取 $x = x_0 + 2t = x(t)$, ~~$u(x(t),0) = (x_0+2t)^2$~~

$$\frac{du}{dt} = \frac{\partial u}{\partial t} + \frac{dx}{dt}\frac{\partial u}{\partial x} = 0 \Rightarrow u(x(t),t) = u(x(0),0) = x_0^2$$

$$\Rightarrow u(x_0+2t,t) = x_0^2$$

$$\Rightarrow u(x,t) = u(x-2t,0) = (x-2t)^2$$

(2) $\partial_t u - \frac{1}{2}\partial_x u = -\frac{1}{2}xu$

選特徵线 $x = x_0 - \frac{1}{2}t$, $\frac{du(x,t)}{dt} = -\frac{1}{2}xu$

$$\Rightarrow \frac{d}{dt}u(x(t),t) = -\frac{1}{2}x(t)u(x(t),t)$$

$$\ln u(x(t),t) - \ln u(x(0),0) = -\frac{1}{2}\int_0^t (x_0 - \frac{1}{2}t)\,dt = -\frac{1}{2}x_0 t + \frac{1}{8}t^2$$

$$\ln u(\underbrace{x(t)}_{x_0-\frac{1}{2}t},t) = -\frac{1}{2}x_0 t + \frac{1}{8}t^2 + \ln(2x_0 e^{\frac{1}{2}x_0^2}) = -\frac{1}{2}x_0 t + \frac{1}{8}t^2 + \ln 2 + \ln x_0 + \frac{1}{2}x_0^2$$

$$\ln u(x,t) = -\frac{1}{2}(x+\frac{1}{2}t)t + \frac{1}{8}t^2 + \ln 2 + \ln(x+\frac{1}{2}t) + \frac{1}{2}(x+\frac{1}{2}t)^2$$

$$= \frac{1}{2}(x+\frac{1}{2}t)(x-\frac{1}{2}t) + \frac{1}{8}t^2 + \ln 2 + \ln(x+\frac{1}{2}t)$$

$$= \frac{1}{2}x^2 + \ln 2 + \ln(x+\frac{1}{2}t) = \frac{1}{2}x^2 + \ln(2x+t)$$

$$u(x,t) = (2x+t)e^{\frac{1}{2}x^2}$$

習題 2.5

證:
$$\Box M_\psi(x,t) = 0,\quad M_\psi(x,0) = 0,\quad \partial_t M_\psi(x,0) = \varphi(x)$$

$$\Rightarrow \Box(\partial_t M_\varphi(x,t)) = 0,\quad \partial_t M_\varphi(x,0) = \varphi(x),\quad \partial_t(\partial_t M_\varphi(x,0)) = 0$$

$$u_1 = \partial_t M_\varphi(x,t)$$

~~$\Box M_f(x,t) = 0,\quad M_f(x,t) = 0,\quad \partial_t M_f(x,0) = f(x)$~~

令 $v(x,t) = \int_0^t M_{f_\tau}(x,t-\tau)\,d\tau$

$$\partial_t v(x,t) = M_{f_\tau}(x,t-\tau)\Big|_{\tau=t} + \int_0^t \partial_t M_{f_\tau}(x,t-\tau)\,d\tau \Rightarrow \partial_t v(x,0) = 0.$$

$$M_{f_\tau}(x,0) = 0$$

$$\partial_{tt} v(x,t) = \partial_t M_{f_\tau}(x,0) + \int_0^t \partial_{tt} M_{f_\tau}(x,t-\tau)\,d\tau$$

$$\partial_{tt} v(x,t) - a^2\partial_{xx} v(x,t) = f(x,t) + \int_0^t \partial_{tt} M_{f_\tau}(x,t-\tau)\,d\tau - a^2\int_0^t \partial_{xx} M_{f_\tau}(t-\tau)\,d\tau = f(x,t)$$ #.

習題 2.6

$$u_2(x,t)=\frac{1}{2a}\left(\int_0^{x+at}\psi(\xi)d\xi-\int_0^{x-at}\psi(\xi)d\xi\right)$$

$$\Rightarrow \partial_t u_2(x,t)=\cancel{\frac{1}{2a}}\frac{1}{2}\left(\psi(x+at)+\psi(x-at)\right)$$

$$u_1=\partial_t M_\varphi(x,t)=\partial_t\cdot\frac{1}{2a}\int_{x-at}^{x+at}\varphi(\xi)d\xi=\frac{1}{2}\left(\varphi(x+at)-\varphi(x-at)\right)$$

$$u_3=\int_0^t M_{f_\tau}(x,t-\tau)d\tau=\int_0^t\frac{1}{2a}\int_{x-a(t-\tau)}^{x+a(t-\tau)}f_\tau(\xi)d\xi\,d\tau=\int_0^t\frac{1}{2a}\int_{x-a(t-\tau)}^{x+a(t-\tau)}f(\xi,\tau)d\xi\,d\tau$$

$$u(x,t)=\frac{1}{2}\left(\varphi(x+at)+\varphi(x-at)\right)+\frac{1}{2a}\int_{x-at}^{x+at}\psi(\xi)d\xi+\frac{1}{2a}\int_0^t\int_{x-a(t-\tau)}^{x+a(t-\tau)}f(\xi,\tau)\,d\xi\,d\tau$$

習題 2.7

(1) $$u(-x,t)=\frac{1}{2}\left(\varphi(-x+at)+\varphi(-x-at)\right)+\frac{1}{2a}\int_{-x-at}^{-x+at}\psi(\xi)d\xi+\frac{1}{2a}\int_0^t\int_{-x-a(t-\tau)}^{-x+a(t-\tau)}f(\xi,\tau)\,d\xi\,d\tau$$

$$=-\frac{1}{2}\left(\varphi(x+at)+\varphi(x-at)\right)-\frac{1}{2a}\int_{x-at}^{x+at}\psi(\xi)d\xi-\frac{1}{2a}\int_0^t\int_{x-a(t-\tau)}^{x+a(t-\tau)}f(\xi,\tau)\,d\xi\,d\tau$$

$$=-u(x,t)$$

(2) $$u(x+L,t)=\frac{1}{2}\left(\varphi(x+L+at)+\varphi(x+L-at)\right)+\frac{1}{2a}\int_{x+L-at}^{x+L+at}\psi(\xi)d\xi+\frac{1}{2a}\int_0^t\int_{x+L-a(t-\tau)}^{x+L+a(t-\tau)}f(\xi,\tau)\,d\xi\,d\tau$$

$$=\frac{1}{2}\left(\varphi(x+at)+\varphi(x-at)\right)+\frac{1}{2a}\int_{x-at}^{x+at}\psi(\xi+L)d\xi+\frac{1}{2a}\int_0^t\int_{x-a(t-\tau)}^{x+a(t-\tau)}f(\xi+L,\tau)\,d\xi\,d\tau$$

$$=\frac{1}{2}\left(\varphi(x+at)+\varphi(x-at)\right)+\frac{1}{2a}\int_{x-at}^{x+at}\psi(\xi)d\xi+\frac{1}{2a}\int_0^t\int_{x-a(t-\tau)}^{x+a(t-\tau)}f(\xi,\tau)\,d\xi\,d\tau$$

$$=u(x,t)$$

習題 2.8

$$\frac{\partial^2(ru)}{\partial t^2}-a^2\frac{\partial^2(ru)}{\partial r^2}=\frac{\partial}{\partial t}\left(r\frac{\partial u}{\partial t}+u\frac{\partial r}{\partial t}\right)-a^2\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}+u\right)$$

$$=r\frac{\partial^2 u}{\partial t^2}+\cancel{\frac{\partial r}{\partial t}\frac{\partial u}{\partial t}}-a^2\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}+u\right)$$

$$r^2=x^2+y^2+z^2$$

$$2r\frac{\partial r}{\partial x}=2x\Rightarrow\frac{\partial r}{\partial x}=\frac{x}{r}$$

$$=r\frac{\partial^2 u}{\partial t^2}-a^2\frac{\partial}{\partial r}\left(r\cdot\left(\frac{\partial u}{\partial x}\cdot\frac{x}{x}+\frac{\partial u}{\partial y}\cdot\frac{y}{y}+\frac{\partial u}{\partial z}\frac{z}{z}\right)+u\right)$$

$$=r\frac{\partial^2 u}{\partial t^2}-a^2\left(\frac{\partial u}{\partial r}\right.$$

习题 2.8

$$r = \sqrt{x^2+y^2+z^2}$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} = \frac{x}{r}\frac{\partial u}{\partial r} \Rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{1}{r}\frac{\partial u}{\partial r} + x\left(-\frac{1}{r^2}\frac{\partial r}{\partial x}\frac{\partial u}{\partial r} + \frac{1}{r}\frac{\partial^2 u}{\partial x \partial r}\right)$$

$$= \frac{r^2 - x^2}{r^3}\frac{\partial u}{\partial r} + \frac{x^2}{r^2}\frac{\partial^2 u}{\partial r^2}$$

$$\Delta u = \frac{\partial^2 u}{\partial r^2} + 2\frac{\partial u}{\partial r}$$

$$\frac{\partial^2 u}{\partial t^2} - a\Delta u = \frac{\partial^2 u}{\partial t^2} - a^2\frac{\partial^2 u}{\partial r^2} - \frac{2a^2}{r}\frac{\partial u}{\partial r} = 0$$

$$\Rightarrow \frac{\partial^2 (ru)}{\partial t^2} - a\frac{\partial^2 (ru)}{\partial r^2} = 0$$